T. A. MARRYSHOW COMMUNITY COLLEGE

DEPARTMENT OF INFORMATION TECHNOLOGY

LAB NAME: LAB 10 WIRELESS MONITORING , WLAN SECURITY & NETSH

COURSE NAME AND NUMBER: DATA SECURITY CONCEPTS CIT245

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TABLE OF CONTENTS

Lab-Objective....2
Pre-Lab Questions (Wireless Basics)....3
Using a Wireless Monitor Tool....4
WLAN Security Information with Vistumbler....5
Configuring a Wireless Router....6
Using Microsoft Windows netsh Commands....7
Evidence & Screenshots....8
Overall Analysis & Reflection....16

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LAB-OBJECTIVE

In this lab, I learned how to see information that Wi-Fi networks broadcast. I used different tools to find nearby networks and see their names, signal strength, and security settings. I also practiced setting up a wireless router safely and used Windows commands to manage Wi-Fi.

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PRE-LAB QUESTIONS (WIRELESS BASICS)

1. **Simple Definitions:**
    - **SSID:** The name of the Wi-Fi network that shows up when you look for connections (like "HomeWiFi").
    - **BSSID:** The unique hardware address (MAC address) of the router or access point.
    - **Channel:** The specific "lane" a Wi-Fi network uses to send and receive data.
    - **RSSI / signal strength:** A number that shows how strong the Wi-Fi signal is. Closer to zero means a stronger signal.
    - **WPA2 / WPA3:** Security methods that lock your Wi-Fi with a password and scramble the data so others can't read it.

2. **2.4 GHz vs 5 GHz:**
    - **GHz:** Goes further through walls but is slower and more crowded (like a busy highway).
    - **5 GHz:** Faster with less interference but doesn't go as far (like a less crowded but shorter road).

3. **Why Use Encryption?**
   Open Wi-Fi (no password) lets anyone nearby see what you do online. They could steal passwords or see your personal information. Encryption keeps your data private.

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USING A WIRELESS MONITOR TOOL

Tool Used: NirSoft WifiInfoView

1. **What it shows:** Network names (SSID), router MAC address (BSSID), signal strength (RSSI), channel number, and security type.

2. **Networks I Found:**

Network Name (Example)	Channel	Band	Signal	Security
My Network	6	2.4	-58 dBm	WPA2
Neighbour 5G	44	5	-72 dBm	WPA2
Guest WiFi	1	2.4	-65 dBm	Open

3. **What I Noticed:** Many networks are on channels 1, 6, and 11. This can cause interference and slow down connections. My own network had the strongest signal, which makes sense since I'm close to it.

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WLAN Security Information with Vistumbler

Tool Used: Vistumbler

1. **Extra Info Vistumbler Shows:** Vistumbler adds graphs that show signal strength over time. It also organizes networks by their security type (like WPA2 or Open), making it easy to see which networks are safe and which are not.

2. **Network Details:**

Field	Information
SSID	N/A
BSSID (MAC)	BO:A7:B9:XX:XX:XX
Channel	6
Signal	80%
Security	WPA2-Personal / AES

3. **Is This Network Secure?**
    - **Good:** It uses WPA2 with AES, which is strong encryption.
    - **Could be better:** It's on 2.4 GHz, so the signal travels far. Someone outside could potentially pick it up.

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CONFIGURING A WIRELESS ROUTER

Tool Used: TRENDnet Online Emulator

1. **Settings I Chose:**

Setting	What I Picked
New SSID	MySecureNetwork
Security Mode	WPA2-PSK
Encryption	AES
Changed Admin Password?	Yes
Disabled WPS?	Yes

2. **Why These Settings Matter:**
    - **Change Admin Password:** Stops strangers from logging into your router and changing your settings.
    - **WPA2 + AES:** Scrambles your Wi-Fi traffic so no one can spy on you.
    - **Disable WPS:** WPS has a security flaw that can let hackers guess your Wi-Fi password.

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USING MICROSOFT WINDOWS NETSH COMMANDS

1. **netsh wlan show interfaces - What it shows:** This command tells me the name of the network I'm connected to, what security type I'm using (like WPA2), and how strong my signal is.

2. **netsh wlan show profiles - What it shows:** This lists every Wi-Fi network my computer has ever connected to and saved. Security professionals can use this to see where a computer has been.

3. **netsh wlan show profile name="[SSID]" - Security Info:** This shows the security details for a saved network, including the authentication type (WPA2-Personal) and cipher (AES). It confirms a security key is saved.

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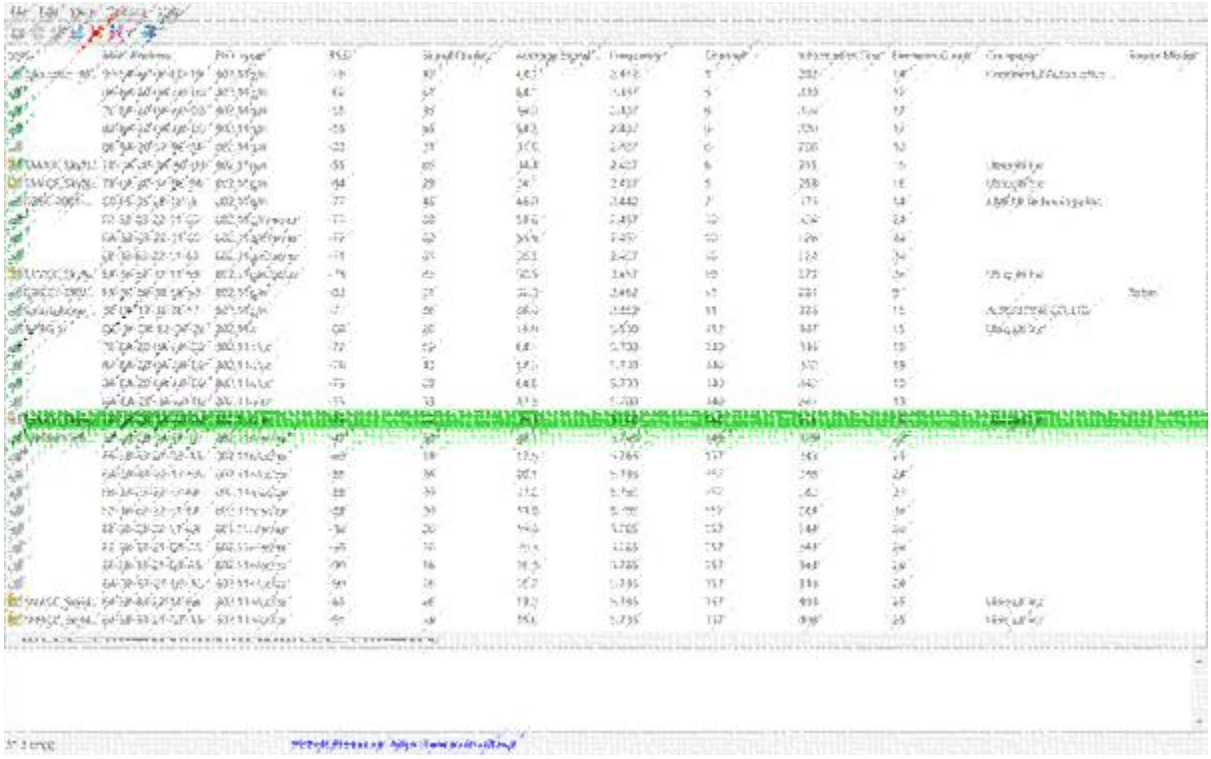
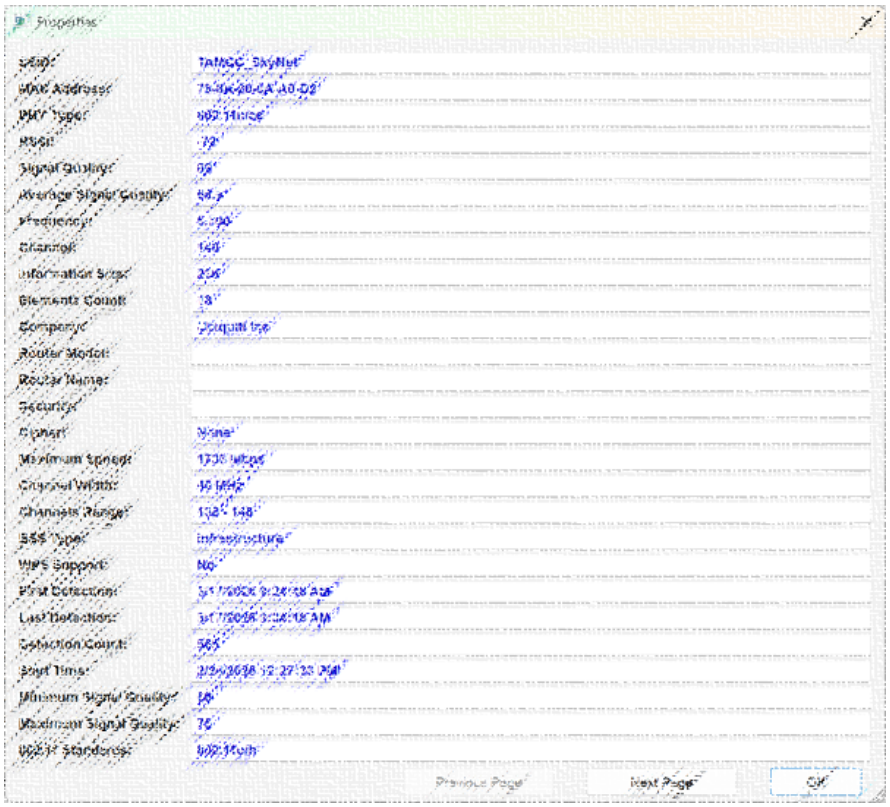
EVIDENCE & SCREENSHOTS

WifiInfoView Main Window



Connected Network Details



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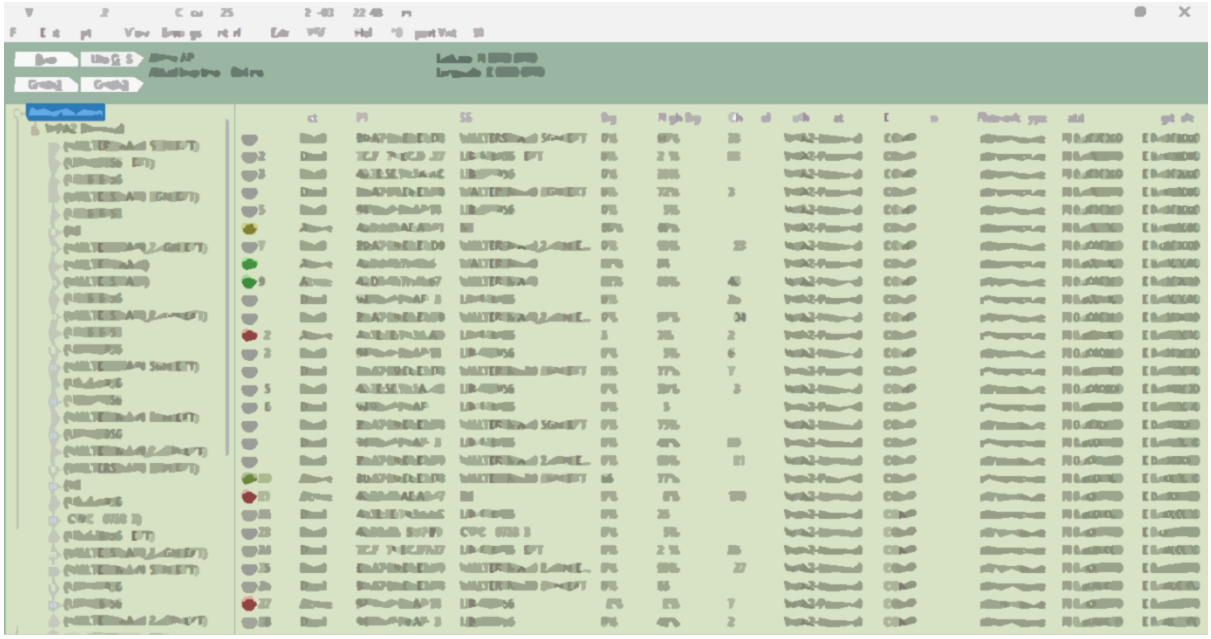
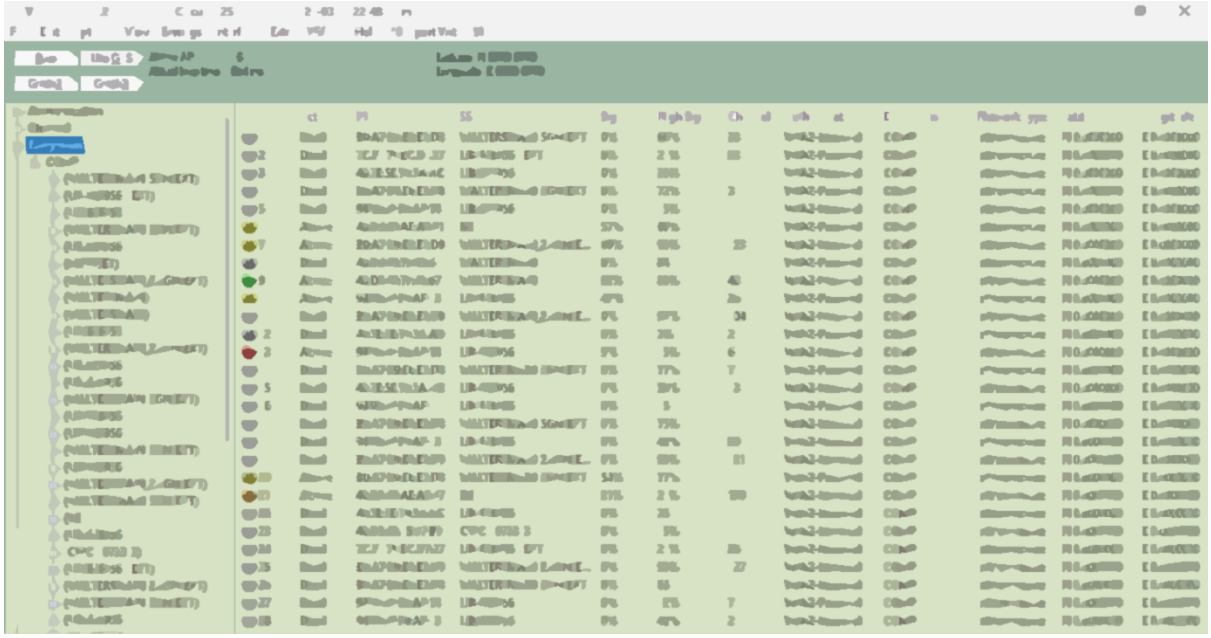
Vistumbler Scan with Authentication Columns



Vistumbler Scan with Encryption Columns



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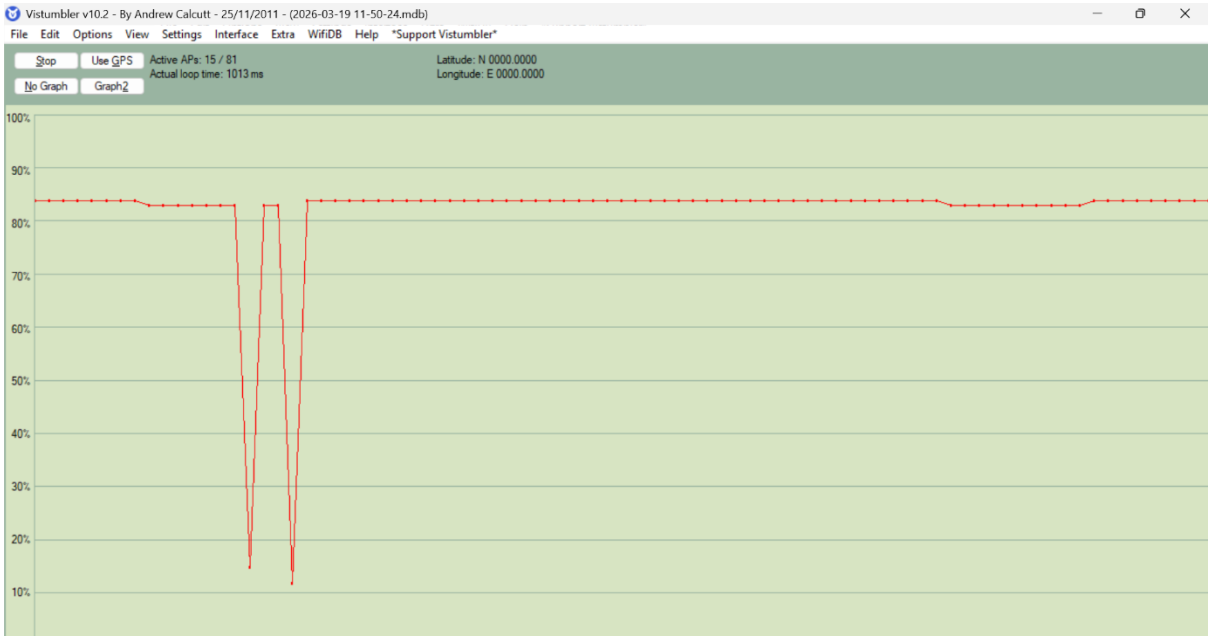
Vistumbler Graph View



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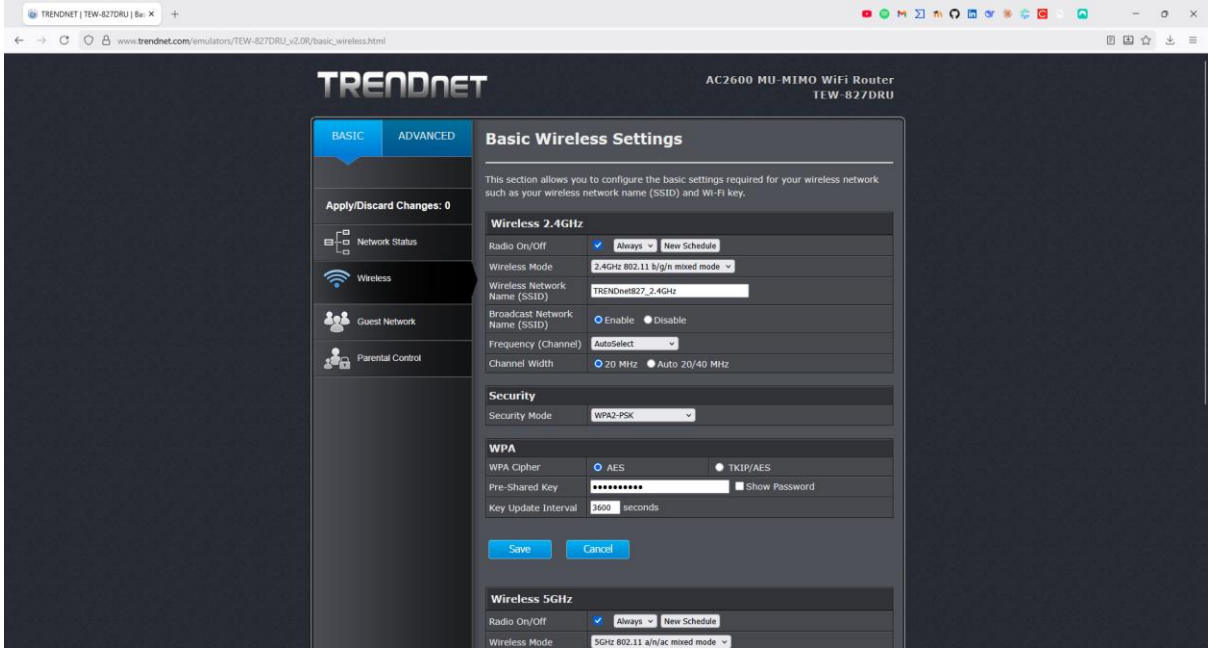
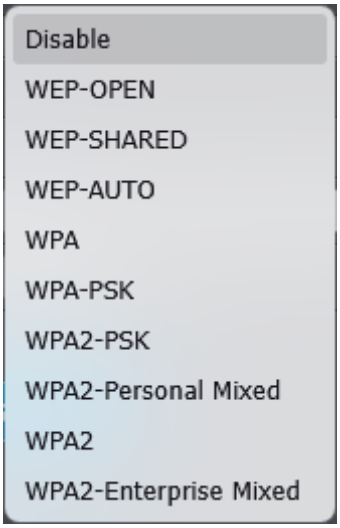
Wireless Security Settings





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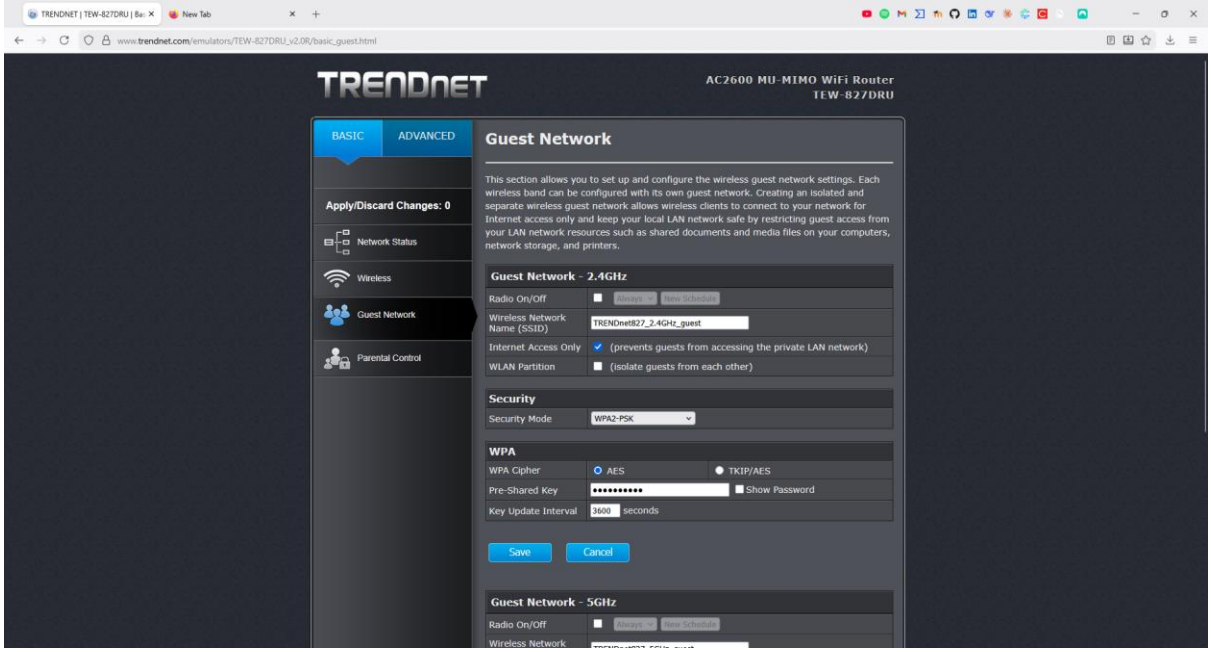
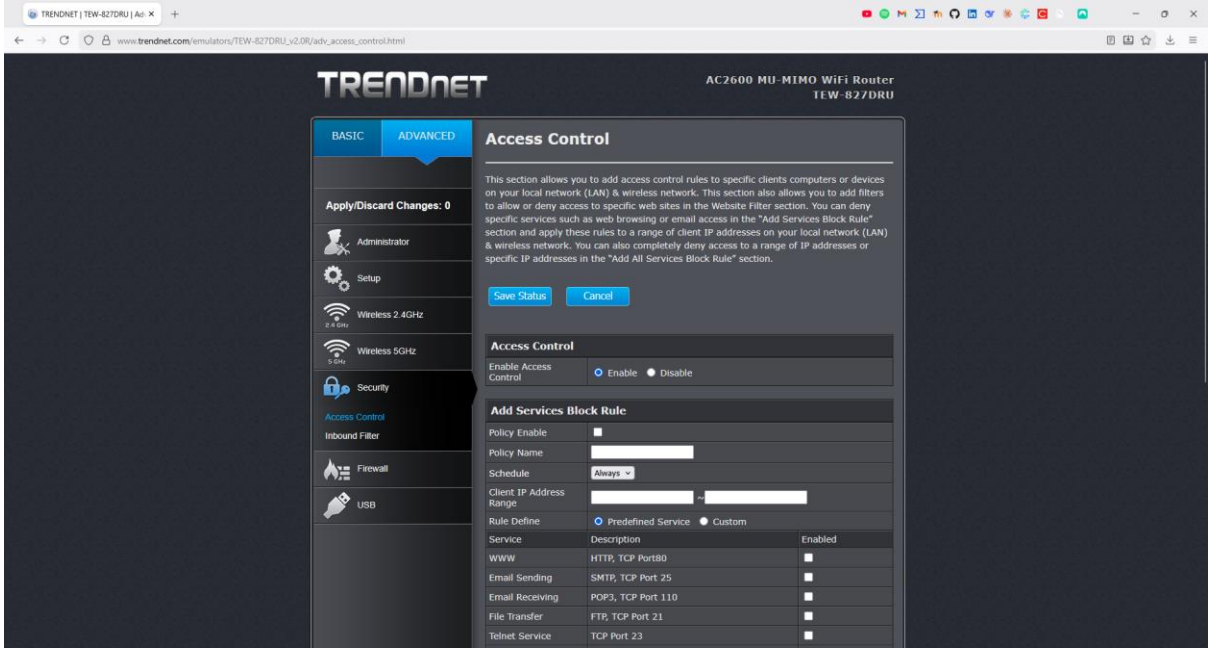
Guest Network Page



Access Control Settings



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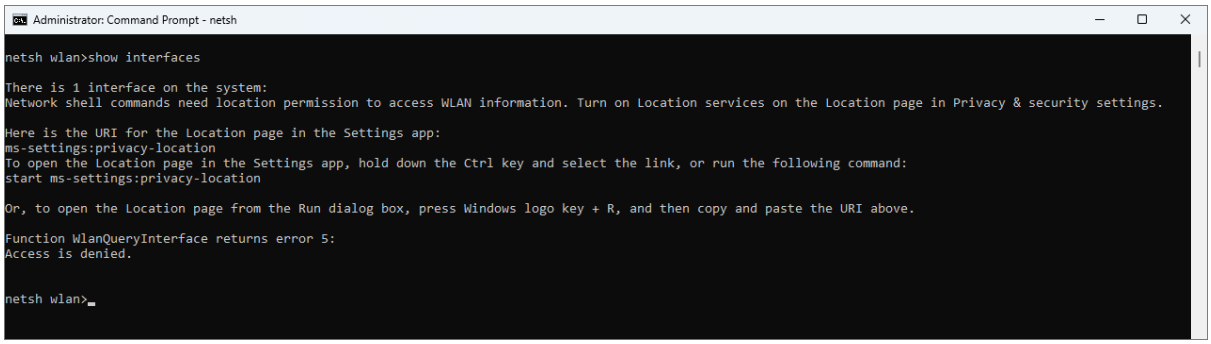
netsh wlan show interfaces



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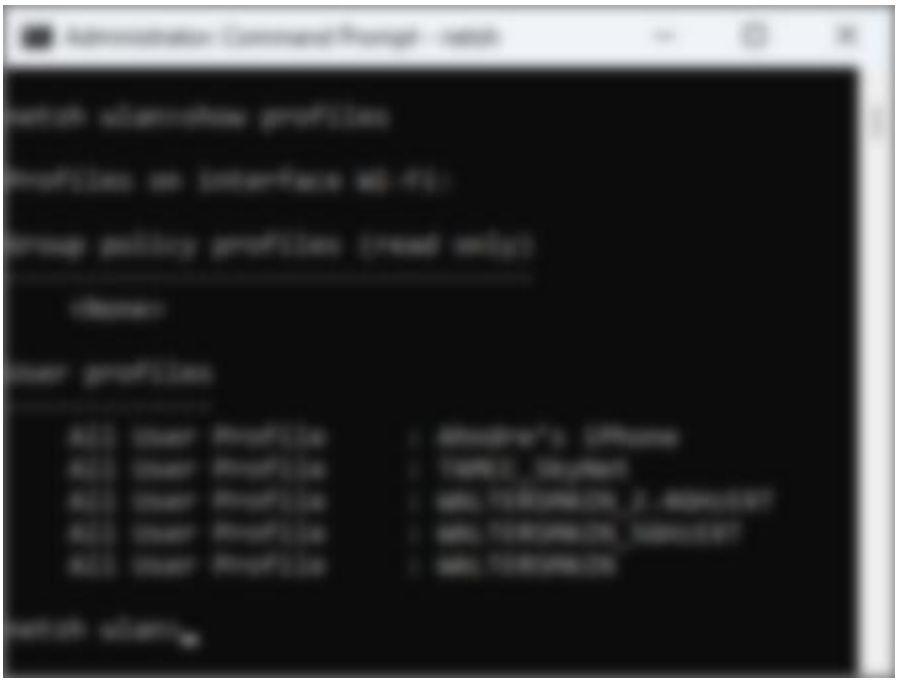
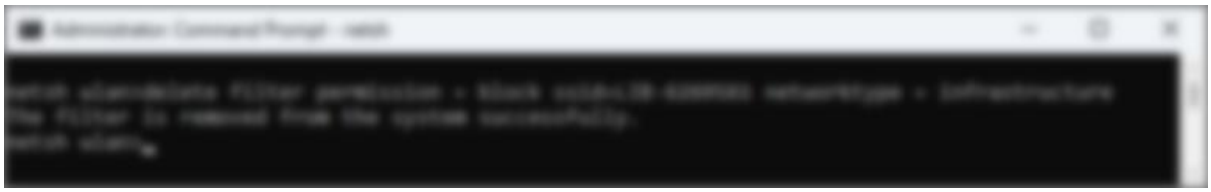
netsh wlan show profiles



netsh add filter block Command

netsh delete filter Command



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WLAN Report in Browser

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OVERALL ANALYSIS & REFLECTION

1. **How Tools Like Vistumbler Help:**
   Security pros can use these tools to find "rogue" access points (unauthorized Wi-Fi) and make sure all company Wi-Fi is using proper security like WPA2. They can also check for signal problems.

2. **Three Router Security Tips:**
   - Change the default admin password.
   - Turn on WPA2 or WPA3 encryption.
   - Turn off WPS.

3. **What I Learned:**
   I learned that Wi-Fi networks broadcast a lot of information that anyone can see. Tools like WifiInfoView and Vistumbler make this easy to find. I also learned how to use netsh commands to manage Wi-Fi from the command line and how to properly secure a home router. At first, I typed some netsh commands wrong, but using the help command fixed it. Next time, I want to try mapping signal strength with GPS.

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